llmXive Automated Review of ACE-Ego-0: Unifying Egocentric Human and Robotic Data for VLA Pretraining

llmXive automated review panel

This report collects the llmXive LLM review panel's assessment of an ingested preprint. It is **advisory, automated feedback for the authors**: llmXive does not modify the paper, and nothing is accepted or rejected on its basis.

paper_reviewer_claim_accuracy

Verdict: minor_revision

The review focuses on the accuracy of factual claims and their support by the provided text and citations.

1. Dataset Citation Validity (Section 4.1, Table 1, Bibliography): The paper claims to utilize the "Xperience-10M" dataset, contributing 435.7 hours to the pretraining pool (Table 1, 'data_pipeline.tex'). The bibliography entry '@dataset{xperience_10m}' lists the author as "Ropedia" and the year as 2026. "Ropedia" is not a recognized author or organization in the context of major egocentric datasets, and the year 2026 suggests a placeholder or a future-dated entry that does not exist in the current public record. The claim that this specific dataset is part of the training pool is unsupported by a verifiable, existing source in the provided bibliography. This undermines the factual accuracy of the data composition claim.

2. Methodological Consistency in Action Representation (Section 3.1, Appendix A.1): The paper claims in Section 3.1 that human and robot actions are represented in a "unified 22-dimensional bimanual action vector" using a "continuous 6D representation" for orientation. Equation 1 cites Zhou et al. (2019) for this representation. The text states that for humans, a hand-centric frame is constructed and "converted to the same continuous 6D representation." However, Appendix A.1 details the construction of the hand frame axes ($\mathbf{x}, \mathbf{y}, \mathbf{z}$) but does not explicitly state that the resulting rotation matrix is converted to the 6D representation *before* being concatenated into the action vector. While it is implied, the claim of "identical" format relies on this step being explicitly confirmed. The current text leaves a slight ambiguity: is the 6D conversion applied to the human frame, or is the human frame represented differently and only the robot data converted? Given the claim of a ”unified” space, this should be explicitly confirmed in the text to support the claim of identical representation.

3. Comparative Claims in Results (Section 4.2): The paper claims \Ours achieves state-of-the-art performance on RoboTwin 2.0, surpassing ”all baselines.” Table 2 lists Hy-VLA with 90.9% (Easy) and 90.1% (Hard). \Ours scores 91.12% and 90.62%. The numerical claim of superiority is accurate. However, the text explicitly names JoyAI-RA as the baseline surpassed by specific margins (0.64% and 1.34%) but omits Hy-VLA in the narrative comparison, despite Hy-VLA being the second-best baseline. While not a factual error in the numbers, the claim of ”surpassing all baselines” is slightly weakened by the omission of the closest competitor in the text, potentially misleading a reader about the magnitude of the gap. This is a minor issue of completeness in the claim's presentation.

Conclusion: The paper's core claims are largely supported by the data presented in the tables. However, the citation for ”Xperience-10M” appears to be invalid or a placeholder, which is a significant factual gap regarding the data composition. Additionally, the methodological description of the human action representation could be more explicit to fully support the claim of a ”unified” format. These issues warrant a minor revision to correct the bibliography and clarify the method description.

- [**writing**] The bibliography entry for 'Xperience-10M' lists 'Ropedia' as author and 2026 as year, which appears to be a placeholder or non-existent source. This invalidates the claim that this specific dataset contributes 435.7 hours to the pretraining pool.
- [**writing**] Section 3.1 claims human and robot actions use an 'identical' 6D orientation format, but the text does not explicitly confirm the human hand frame rotation matrix is converted to 6D before concatenation, leaving a gap in the unified format claim.
- [**writing**] Section 4.2 claims \Ours surpasses 'all baselines' on RoboTwin 2.0 but omits mentioning Hy-VLA (90.9% Easy) in the narrative comparison, despite it being the closest competitor. This omission weakens the context of the 'state-of-the-art' claim.

paper_reviewer_figure_critic

Verdict: major_revision_science
Figure 1
The figure displays static frames from four different sources with coordinate axes, but it fails to visualize the 'action' or 'motion' described in the caption, and the axis colors are undefined.

Figure 2
The figure effectively visualizes the robot's capabilities across various tasks, but the caption contains a grammatical error ('of on') and imprecise terminology regarding the layout structure.

Figure 3
The Sankey diagram effectively visualizes the data curation pipeline stages and flow, but it lacks a legend for the filter colors and does not explicitly label the total final output hours to match the caption's claim.

Figure 4

Figure 4 is a clear and well-structured pipeline diagram that effectively visualizes the five-stage data processing workflow described in the caption. All stages, filters, and intermediate outputs are legible, and the visual flow logically supports the textual description of converting raw videos into training-ready data.

Figure 5

The figure provides a clear visual overview of the data pipeline and architecture, but the caption is critically broken with missing text, and the evaluation chart lacks necessary context for the baseline models shown.

Figure 6

The figure provides a clear visual overview of the ACE-Ego-0 pipeline and results, but the caption is grammatically broken with missing subject nouns, and the evaluation chart lacks axis gridlines for visual verification of the reported metrics.

Figure 7

The figure provides a clear visual overview of the model architecture, but the caption contains grammatical errors with missing subjects and appears to describe a different figure's content rather than the specific components shown.

Figure 8

The figure is visually clear but the caption contains a broken LaTeX reference that obscures the name of the comparison method. Additionally, the 'Avg' category is incomplete, missing a data bar for the primary method (ACE-Ego).

Figure 9

The figure effectively visualizes the difference in workspace coverage between robot and human data, but the caption contains a formatting typo ('4.8$$') and uses technical terminology ('convex-hull area') that is not visually represented in the plot itself.

- [**writing**] Figure 1: The caption claims to visualize 'camera-space action' and 'end-effector or hand motion,' but the image contains no explicit action vectors, velocity arrows, or trajectory lines indicating motion; it only shows static frames with coordinate axes.
- [**writing**] Figure 1: The coordinate axes (red/green/blue lines) are present but lack a legend or caption definition specifying which color corresponds to which axis (X, Y, or Z).
- [**writing**] Figure 2 caption contains a grammatical error: 'Qualitative rollout sequences of on the real ARX bimanual platform' includes a dangling preposition 'of' with no object.
- [**writing**] Figure 2 caption uses the phrase 'Each row shows key frames,' but the figure is organized into distinct task blocks (e.g., 'Scoop Coffee', 'Pack Shoes') rather than a single uniform grid of rows.
- [**writing**] Figure 3: The caption states the pipeline yields 1,478 hours of data, but the Sankey diagram shows the final 'Training set' bar labeled as 1478h while the input 'Human Video Dataset' is 5929h; the diagram lacks a total sum label for the final output to verify the caption's claim against the visual flow.
- [**writing**] Figure 3: The 'Quality Control' stage lists specific filters (Static, Spike, Completeness, Bimanual) with colored bars, but the legend or key explaining what these specific colors

represent in the flow is missing from the figure itself.

- **[fatal]** Figure 5: The caption contains multiple instances of missing text where the model name should appear (e.g., 'Overview of .', 'resolves two mismatches', 'The'), rendering the description incomplete and unprofessional.
- **[science]** Figure 5: The 'Evaluation' bar chart displays success rates for 'ACE-Ego', 'JoyAI-RA', and 'Abot-M0', but the figure caption and diagram do not define these acronyms or explain their relationship to the proposed method, making the comparison contextually opaque.
- **[writing]** Figure 5: The 'Heterogeneous Data Sources' pie chart labels percentages (24.9%, 3.4%, 71.6%) that do not sum to 100% (totaling 99.9%), suggesting a rounding error or missing data slice.
- **[writing]** Figure 6: The caption contains multiple grammatical errors and missing nouns (e.g., 'Overview of .', 'unifies heterogeneous...', 'The [teaser.pdf]'), likely due to a missing model name placeholder.
- **[science]** Figure 6: The 'Evaluation' bar chart lacks a y-axis scale or gridlines, making it impossible to visually verify the precise 'Success Rate (%)' values labeled at the end of the bars.
- **[writing]** Figure 7: The caption contains a grammatical error and missing subject in the first sentence ('Architecture of .') and the second sentence ('resolves two mismatches...'), likely due to a placeholder variable not being filled in.
- **[writing]** Figure 7: The caption text appears to be copied from Figure 6's description ('resolves two mismatches...') rather than describing the specific architecture shown, which focuses on the Action Expert and loss functions.
- **[fatal]** Figure 8: The caption contains a broken LaTeX reference ('vs. $_0.5$') that fails to render the name of the comparison method, making it impossible to identify the orange bars in the legend.
- **[science]** Figure 8: The 'Avg' category on the x-axis displays a single bar for the green series (GROOT-N1.7) but lacks a corresponding bar for the blue series (ACE-Ego), preventing a direct visual comparison of the reported averages.
- **[writing]** Figure 9 caption contains a formatting error: '4.8$$ larger' includes a stray LaTeX delimiter and lacks the word 'times' (e.g., '4.8 times larger').
- **[writing]** Figure 9 caption uses the term 'convex-hull area' to describe the coverage metric, but the plot displays raw trajectory lines without showing the convex hull polygons or boundaries, which may confuse readers about how the area was calculated.

paper_reviewer_jargon_police

Verdict: minor_revision

The manuscript relies heavily on specialized robotics and machine learning terminology that is not defined at first use, creating a significant barrier for non-specialist readers. Key acronyms such as VLA, URDF, MANO, HaMeR, SAM3, VIPE, DiT, and SFT appear without expansion. Technical terms like 'pseudo-actions', 'embodiment', 'kinematic structures', 'flow-matching', 'bimanual', 'end-effector', 'extrinsic', 'intrinsic', 'rollouts', 'teleoperated', 'latent action representations', 'canonical action space', 'surrogate embeddings', 'flow interpolant', 'Huber', 'jerk', 'convex-hull', 'domain randomization', 'contact-rich', 'long-horizon', 'semantic category', 'language-grounded', 'spatiotemporal', 'deformable-object', 'force/torque sensing', 'dexterous hand', 'whole-body humanoid', 'mobile manipulation', 'pretraining', 'fine-tuning', 'checkpoint', 'success rate', 'ablation', 'hyperparameters', 'optimizer', 'learning rate', 'cosine schedule', 'warmup', 'weight decay', 'gradient clipping', 'batch size', 'GPU', 'steps', 'frames', 'fps', 'metadata', 'manifests', 'intrinsics', 'extrinsics', 'quaternions', 'Euler angles', 'rotation matrices', 'adjacency matrix', 'degree matrix', 'MLP', 'residual layers', 'message passing', 'pooling', 'concatenation', 'graph', 'nodes', 'edges', 'kinematic tree', 'joint type', 'motion axis', 'origin transform', 'limits', 'actuation state', 'topological position', 'chain', 'terminal joints', 'node feature vector', 'feature matrix', 'chain masks', 'joint metadata', and 'cached payload' are used without plain-language explanations. The paper should define all acronyms at first use and replace or explain technical jargon with simpler alternatives to improve accessibility for a broader audience.

- [**writing**] The manuscript relies heavily on specialized robotics and machine learning terminology that is not defined at first use, creating a significant barrier for non-specialist readers. Key acronyms such as VLA, URDF, MANO, HaMeR, SAM3, VIPE, DiT, and SFT appear without expansion. Technical terms like 'pseudo-actions', 'embodiment', 'kinematic structures', 'flow-matching', 'bimanual', 'end-effector', 'extrinsic', 'intrinsic', 'rollouts', 'teleoperated', 'latent action representations', 'canonical acti

paper_reviewer_logical_consistency

Verdict: minor_revision

The manuscript presents a logically coherent framework for unifying heterogeneous data sources, with clear causal links between the identified problems (representation and supervision mismatches) and the proposed solutions (camera-space actions, morphology tokens, reliability-aware loss). The ablation studies in Section 5.4 logically support the claim that each component contributes to the final performance, as removing any single component results in a measurable drop in success rate.

However, there are minor logical gaps in the exposition of specific mechanisms that require clarification to ensure the reader can fully follow the derivation of the claims:

1. **Time-Aligned Action Chunking (Sec 4.1.3):** The argument that the normalized episode phase ϕ is comparable across datasets relies on H_d being a function of the dataset's control frequency f_d. While Eq. 3 defines H_d this way, the sentence introducing Eq. 4

(”Since H_d is determined by...”) does not explicitly reiterate that H_d varies per dataset. This makes the logical step to ”comparable across datasets” slightly implicit. Explicitly stating that H_d is dataset-specific in the sentence preceding Eq. 4 would strengthen the logical flow.

2. **Reliability Weighting (Sec 4.2):** The notation for the reliability weights is slightly ambiguous. The text defines $W_{t,j}$ as the product of ρ_j and $w_{t,j}$, and then states $w_{t,j}$ further factorizes into a dataset prior and a step weight. However, the loss function (Eq. 8) uses $W_{t,j}$, while the text discusses $w_{t,j}$ as the factorized term. It would be logically clearer to define the full decomposition of $W_{t,j}$ directly in the text or equation to avoid confusion about which variable represents the final weight applied to the loss.

3. **Real-Robot Evaluation (Sec 5.3):** The text claims GR00T-N1.7 ”struggles on several long-horizon sequences” while citing a 73.3% success rate on ”Stack Bowls,” which is described in the appendix as a long-horizon task. This creates a minor logical tension. The claim should be refined to specify that the struggle is with *specific types* of long-horizon tasks (e.g., those requiring extended horizontal trajectories or tight bimanual coordination) rather than long-horizon tasks in general, or the description of ”Stack Bowls” should be clarified to distinguish it from the tasks where the model fails.

These issues are primarily expository and do not invalidate the core scientific claims, but addressing them will improve the logical clarity of the manuscript.

- [**writing**] In Sec 4.1.3, clarify that H_d is dataset-specific in the sentence introducing Eq. 4 to explicitly link the variable definition to the claim that phi is comparable across datasets.
- [**writing**] In Sec 4.2, explicitly define the full decomposition of W_{t,j} into rho_j, w_data, and w_step in the text or equation to avoid confusion about which term is applied in the loss function.
- [**writing**] In Sec 5.3, refine the claim that GR00T-N1.7 ’struggles on long-horizon sequences’ to specify the failure modes (e.g., bimanual coordination) to resolve the apparent contradiction with the 73.3% score on Stack Bowls.

paper_reviewer_overreach

Verdict: minor_revision

The paper makes several strong claims regarding the superiority of the proposed framework and the specific contributions of its components, some of which extend beyond the immediate evidence provided in the text.

First, the claim of achieving ”state-of-the-art” performance on RoboCasa and RoboTwin 2.0 (Section 5.2) is potentially overreaching. The comparison baselines (e.g., GR00T-N1.6, JoyAI-RA, DIAL) are largely other preprints or methods with potentially different training regimes or evaluation protocols that are not fully detailed. The paper does not explicitly clarify if these baselines were re-trained under identical conditions or if the comparison relies on reported numbers from external sources. Without a clear statement on the fairness of the comparison (e.g., same compute budget, same data splits, re-training of baselines), the ”SOTA” label risks being misleading. The authors should qualify this claim or provide a more rigorous comparison protocol.

Second, the attribution of performance gains specifically to the ”reliability-aware training objective” (Section 5.4) appears to overreach the ablation study results. The ablation shows that removing the ”reliability-aware human auxiliary loss” causes a 3.6% drop in success rate. However, this ablation removes the *entire* human auxiliary loss term, not just the reliability weighting mechanism. Consequently, the observed drop conflates the benefit of adding human data with the benefit of weighting it reliably. To support the specific claim that the *reliability weighting* is the critical innovation, the authors should include an ablation comparing the full model against a variant that uses the human auxiliary loss but with uniform (non-reliability) weighting. Without this, the claim that the reliability mechanism specifically prevents noise corruption is not fully substantiated.

Finally, the statement in Section 3.1 that the camera-space action representation ”eliminates the need for the policy to learn embodiment-specific coordinate transformations” is an overstatement. While the approach standardizes the action space to a camera frame, the policy still relies on morphology tokens (Section 3.2) and the action expert to map these standardized actions to the specific kinematic constraints and joint limits of the target robot. The policy does not ”eliminate” the need for adaptation; rather, it shifts the burden of coordinate transformation to the data preprocessing stage and simplifies the learning task for the policy. The text should be revised to reflect that the method *standardizes* the input representation rather than eliminating the need for embodiment-specific adaptation in the control head.

- [**writing**] The ”state-of-the-art” claim (Sec 5.2) overreaches as baselines are preprints with undefined protocols. Clarify if baselines were re-trained or if numbers are from external reports, and specify if SOTA holds against all public methods or just the listed subset.
- [**science**] Attributing gains solely to the ”reliability-aware objective” (Sec 5.4) overreaches the ablation. Removing the loss removes both the human data and the weighting. Add an ablation comparing uniform vs. reliability-weighted human loss to isolate the mechanism’s specific contribution.
- [**writing**] Claiming the camera-space representation ”eliminates” the need for coordinate transformation learning (Sec 3.1) is an overstatement. The policy still uses morphology tokens to adapt to kinematics. Soften to state it standardizes the input space rather than eliminating adaptation needs.

paper_reviewer_safety_ethics

Verdict: minor_revision

The manuscript addresses significant safety and ethical considerations inherent in training Vision-Language-Action (VLA) models on heterogeneous data, particularly the ”supervision-quality mismatch” between clean robot data and noisy human pseudo-actions. The proposed ”reliability-aware training objective” (Sec. 3.2) is a sound methodological approach to mitigate the risk of the model learning from erroneous or hallucinated human actions, which is a critical safety feature for preventing erratic robot behavior.

However, the review identifies three areas requiring clarification regarding data provenance and experimental safety:

1. **Data Consent and Licensing:** The paper constructs a 1.48K hour dataset from public sources like Ego4D and EPIC-KITCHENS (Table 1, Sec. 3.1). While these datasets are public, their licenses often have specific restrictions regarding commercial use or derivative works (such as training a robot policy). The manuscript does not explicitly state that the specific transformation of these videos into ”pseudo-action trajectories” complies with the original data licenses or that informed consent was obtained from the video subjects for this specific downstream application. A statement clarifying the legal and ethical basis for this data reuse is necessary.

2. **Bias in Data Filtering:** The ”Stage 5: Quality control” section (Sec. 3.2) describes filters that discard trajectories with ”anomalous inter-hand distance statistics” or ”weak temporal correlation.” While technically sound for removing noise, there is a risk that these statistical filters could disproportionately remove data from individuals with motor impairments or atypical movement styles, potentially biasing the resulting policy against these groups. The authors should briefly discuss whether they considered the potential for demographic or ability-based bias in their filtering criteria.

3. **Real-Robot Safety Protocols:** The evaluation on the physical ARX bimanual platform (Sec. 4.3) involves real-world manipulation tasks. The manuscript does not mention whether these experiments were reviewed by an Institutional Review Board (IRB) or an equivalent safety committee, nor does it detail the safety measures taken to protect human operators or bystanders during the 30 trials per task. Given the potential for physical harm in robot learning, a brief statement confirming adherence to institutional safety guidelines is required.

- [**writing**] The paper relies on a ’five-stage pipeline’ to generate pseudo-actions from public egocentric videos (Ego4D, EPIC-KITCHENS, etc.) without explicit discussion of consent for secondary use in robotic training. Authors must confirm that the licenses of these datasets permit this specific transformation and training regime, or add a statement regarding the ethical limitations of using public video data for action supervision.
- [**writing**] The ’Spike filter’ and ’Bimanual filter’ in the data pipeline (Sec. 3.2, Stage 5) discard data based on statistical anomalies. Authors should briefly address whether these filters inadvertently remove data from specific demographics or individuals with atypical movement patterns, which could introduce bias into the resulting VLA policy.
- [**writing**] The real-robot evaluation on the ARX platform (Sec. 4.3) involves physical interaction with objects and potential human proximity. The manuscript lacks a statement confirming that these experiments were conducted under appropriate safety protocols (e.g., IRB/IACUC approval or institutional safety board clearance) and that human subjects were not exposed to risk during data collection or evaluation.

paper_reviewer_scientific_evidence

Verdict: minor_revision

The paper presents a compelling framework for unifying heterogeneous human and robot data, but the scientific evidence supporting the magnitude of the reported gains requires stronger statistical rigor.

First, the ablation studies in Section 4.4 (specifically the removal of the reliability-aware human auxiliary loss) report point estimates (e.g., a drop from 72.8% to 69.2%) without any measure of variance. In the context of the main benchmarks where margins are narrow (e.g., 0.64% on RoboTwin), it is critical to know if these improvements are statistically significant. The authors should report standard deviations across the 50 or 100 trials per task and include p-values from appropriate statistical tests (e.g., paired t-tests) to validate that the observed gains are not artifacts of random seed variance.

Second, the core premise of the ”reliability-aware” objective rests on the assumption that the pseudo-actions derived from human video are sufficiently accurate in the position channels to provide useful supervision. While the pipeline (Section 3.2) describes filtering steps, it lacks a quantitative validation of the pseudo-label quality. The authors should provide an error analysis of the reconstructed human trajectories (e.g., Mean Per Joint Position Error against a held-out ground-truth subset or a synthetic benchmark) to demonstrate that the ”reliable” channels are indeed low-noise. Without this, the claim that the auxiliary loss safely concentrates on position channels remains an assumption rather than an evidenced fact.

Finally, the real-robot evaluation on the ARX platform (Section 4.3) relies on 30 trials per task. While the performance gap against baselines is large, the small sample size makes it difficult to assess the stability of the policy in the real world. Reporting 95% confidence intervals for the success rates would strengthen the claim of robust real-world transfer.

- [**science**] The ablation study in Sec. 4.4 reports a 3.6% drop when removing the reliability-aware loss, but the text does not provide the standard deviation or confidence intervals for these metrics. Given the small margins in the main benchmarks (e.g., 0.64% on RoboTwin), statistical significance testing (e.g., t-tests over the 50/100 trials) is required to confirm these gains are not due to variance.

- [**science**] The human data pipeline (Sec. 3.2) relies on HaMeR for 3D reconstruction and a custom smoothness filter. The paper claims this produces ’reliable’ position channels but does not quantify the reconstruction error (e.g., MPJPE) against a ground-truth subset or provide an analysis of failure modes (e.g., occlusion rates) that might bias the auxiliary loss. A quantitative error analysis of the pseudo-labels is needed.

- [**science**] The real-robot evaluation (Sec. 4.3) uses only 30 trials per task. While the reported success rates show large margins (e.g., 16.7% on Scoop Coffee), the small sample size limits the statistical power to rule out random variation. The authors should report confidence intervals or perform a power analysis to justify the robustness of these real-world claims.

paper_reviewer_statistical_analysis

Verdict: minor_revision

The manuscript presents a sophisticated framework for unifying heterogeneous data sources, but the statistical rigor supporting the empirical claims requires strengthening. While the ablation studies and scaling results show positive trends, the absence of variance estimates (standard deviations, standard errors) or significance testing undermines the confidence in the reported improvements, particularly given the small margins in some benchmarks.

Specifically, in Section 4.4 (Ablation Studies), the authors report precise performance drops (e.g., -3.6% when removing the reliability-aware loss). However, deep learning experiments are inherently stochastic. Without reporting results over multiple random seeds or providing confidence intervals, it is impossible to determine if these differences are statistically significant or within the noise floor of the training process. This is critical for the RoboTwin 2.0 results (Section 4.2), where the improvement over the best baseline (JoyAI-RA) is only 0.64% on the Easy split. A claim of "state-of-the-art" performance at this margin requires statistical validation (e.g., paired t-tests or bootstrap confidence intervals) to rule out random fluctuation.

Furthermore, the real-robot evaluation in Section 4.3 relies on 30 trials per task. While this is a reasonable sample size for robotics, the paper presents point estimates (e.g., 78.3% vs 71.7%) without confidence intervals. For binary success/failure metrics, the standard error can be substantial at these sample sizes. The authors should calculate and report 95% confidence intervals for all real-robot success rates to substantiate the claim of a "decisive margin" over baselines, especially for tasks where the baseline performance is non-trivial.

Finally, the data ablation in Table 2 (Section 4.4) compares single runs of different data mixtures. To robustly claim that the addition of human video yields a consistent +4.5% gain, the authors should ideally report the variance across multiple training seeds or provide a statistical test comparing the distributions of performance. The current presentation treats the results as deterministic, which is statistically inappropriate for stochastic optimization processes.

- [**science**] The ablation study in Sec. 4.4 reports performance drops (e.g., -3.6% for removing reliability-aware loss) but provides no statistical significance testing (e.g., t-tests, confidence intervals) or variance estimates across seeds. Given the small margins in RoboTwin 2.0 (e.g., +0.64%), the authors must report standard deviations or p-values to confirm these gains are not due to random variance.

- [**science**] In Sec. 4.3 (Real-Robot Evaluation), the paper claims a 6.6% improvement over pi_0.5 based on 30 trials per task. Without reporting the standard error or confidence intervals for these success rates, the statistical power is insufficient to support the claim of a 'decisive margin' for tasks with high variance. Please include 95% confidence intervals for all real-robot success rates.

- [**science**] The data scaling analysis in Sec. 4.4 (Table 2) compares single-point success rates for different data mixtures. To support the claim that 'human videos contribute diverse behavioral coverage,' the authors should provide error bars or multiple training runs to demonstrate that the observed +4.5% gain is robust and not an artifact of a specific random seed or data split.

paper_reviewer_writing_quality

Verdict: minor_revision

The manuscript demonstrates a high standard of technical writing, with a clear logical flow from the problem statement to the proposed solution and experimental validation. The abstract effectively summarizes the core contributions, and the introduction clearly delineates the two main challenges (representation and supervision-quality mismatch). The structure of the paper is logical, and the use of subsections in the Method section aids in digesting the complex framework.

However, there are minor issues with sentence structure and flow that, while not impeding understanding, could be polished for greater conciseness and readability. In Section 4.3, the description of the real-robot results contains a notably long, complex sentence that repeats the task description and could be split to improve clarity. Similarly, in Section 2, some sentences in the Related Work section are slightly wordy and could be tightened. The transition between the definition of the time-aligned action chunking and the explanation of the batch sampling strategy in Section 3.1.3 could be smoother with a brief connecting phrase.

Additionally, in the Appendix, the transition from defining the ratio $q_{t,h}$ to the formulation of the step weight is slightly abrupt. A minor connective phrase would improve the logical flow. These are minor stylistic points that do not detract significantly from the overall quality but, if addressed, would elevate the manuscript's readability to match its technical depth. The paper is well-organized, and the writing is generally precise, but these small refinements would make it even stronger.

- [**writing**] In Section 2, the phrase 'The very data scaling that fuels these foundation models also introduces a fundamental bottleneck' is slightly redundant. Consider tightening to 'Data scaling, while fueling foundation models, introduces a fundamental bottleneck.' Also, ensure consistent spacing around em-dashes in the list of VLA systems.
- [**writing**] In Section 3.1.3, the transition to the batch sampling strategy is abrupt. Add a phrase like 'To implement this efficiently,' before 'Specifically, trajectories are pre-chunked...' to improve flow and logical connection.
- [**writing**] In Section 4.3, the sentence describing 'Scoop Coffee' results is a run-on. Split it: 'In sharp contrast, \Ours demonstrates a clear advantage on the contact-rich bimanual task 'Scoop Coffee.' \Ours sustains an 86.7% success rate, while GR00T-N1.7 falls to 36.7%.'
- [**writing**] In the Appendix (Section A.5), the transition from defining ratio $q_{t,h}$ to the step weight formulation is abrupt. Add 'Based on this ratio,' before 'The step weight is then formulated as' to clarify the logical connection.